

IMPLEMENTATION ROADMAP

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Version 2.0

Document Purpose

This document defines the complete execution sequence for the project.

The purpose of the roadmap is to transform the scientific plan into a repeatable development process.

This document defines:

- Development phases
- Build sequence
- Validation sequence
- Dependencies
- Completion criteria
- Phase gates
- Deliverables
- Expected outputs

The roadmap is designed to minimize rework, reduce debugging complexity, and ensure that each scientific result is built on validated foundations.

Development Philosophy

The project follows a staged development approach.

Each phase follows the sequence:

Build



Validate



Freeze



Continue

A phase is considered complete only when:

Implementation Complete

Validation Complete

Outputs Verified

Documentation Updated

At that point the phase is frozen and future work builds on the validated outputs.

Master Development Sequence

The complete project development flow is:

Phase 1

Infrastructure



Phase 2

Dataset Framework



Phase 3

Sparsity Framework



Phase 4

Experiment 1
Antisymmetry



Phase 5
Experiment 2
Epistasis



Phase 6
Integrated Reliability Analysis



Phase 7
Reliability Framework



Phase 8
TKT Mutation Landscape



Phase 9
Industrial Fitness Framework



Phase 10
Figures & Publication Assets

PHASE 1

Infrastructure Foundation

Objective

Create the project foundation.

This phase contains no scientific calculations.

Its purpose is to build the software environment required by all later phases.

Deliverables

Repository Structure

Configuration Framework

Logging Framework

Dataset Registry

Validation Framework

Experiment Tracking

Unit Testing Framework

Utilities

Required Directories

configs/

data/

docs/

notebooks/

outputs/

reports/

src/

tests/

Completion Criteria

The phase is complete when:

System Starts Correctly

Configuration Loads

Logging Works

Tests Execute

Dataset Registry Loads

Phase Gate

Infrastructure Validated

before entering Phase 2.

PHASE 2

Dataset Framework

Objective

Build all dataset ingestion and validation systems.

No scientific calculations occur during this phase.

Deliverables

Megascale-D Import

Sequence Import

Structure Import

Dataset Validation

Schema Validation

Version Tracking

Outputs

Validated Source Datasets

including:

megascale_d.parquet

protein_sequences.parquet

protein_structures.parquet

Completion Criteria

Every source dataset must pass:

Schema Validation

Primary Key Validation

Range Validation

Missing Value Validation

Phase Gate

All Source Data Validated

before entering Phase 3.

PHASE 3

Sparsity Framework

Objective

Implement the complete sparsity framework.

This is the first major scientific phase.

Deliverables

Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity

Combined Sparsity

Output Datasets

empirical_sparsity.parquet

evolutionary_sparsity.parquet

structural_sparsity.parquet

combined_sparsity.parquet

Validation Requirements

Verify:

All Scores Between 0 and 1

No Missing Values

Distribution Reasonable

All Merges Successful

Completion Criteria

combined_sparsity.parquet

,

generated successfully.

Phase Gate

Combined Sparsity calculations are frozen before Experiment 1 begins.

PHASE 4

Experiment 1

Sparsity and Thermodynamic Inconsistency

Objective

Determine whether sparsity predicts antisymmetry error.

Deliverables

Forward Predictions

Reverse Predictions

Antisymmetry Error Calculation

Statistical Analysis

Figure Generation

Outputs

antisymmetry_predictions.parquet

antisymmetry_results.parquet

Required Analyses

Correlation

Regression

Quantile Analysis

Distribution Analysis

Primary Figure

Combined Sparsity

vs

Antisymmetry Error

Completion Criteria

Positive, negative, or null results are all acceptable.

The requirement is:

Analysis Completed

Results Verified

Figures Generated

Phase Gate

Experiment 1 outputs frozen before Experiment 2 begins.

PHASE 5

Experiment 2

Sparsity and Prediction Failure

Objective

Determine whether sparsity predicts epistasis error.

Deliverables

Epistasis Calculations

Prediction Comparisons

Error Analysis

Validation Figures

Outputs

epistasis_results.parquet

Required Analyses

Correlation

Regression

Error Distributions

Stratified Comparisons

Primary Figure

Combined Sparsity

vs

Epistasis Error

Completion Criteria

epistasis_results.parquet

produced and validated.

Phase Gate

Experiment 2 frozen before integrated analysis begins.

PHASE 6

Integrated Reliability Analysis

Objective

Determine whether a unified reliability model emerges from the data.

Core Question

Does the following chain exist?

Combined Sparsity



Antisymmetry Error



Epistasis Error

Deliverables

Integrated Statistics

Mediation Analysis

Integrated Dataset

Integrated Figures

Outputs

integrated_reliability_analysis.parquet

Primary Figure

Sparsity



Antisymmetry



Epistasis

Completion Criteria

Integrated relationship successfully evaluated.

Phase Gate

Integrated analysis frozen before Reliability Framework creation.

PHASE 7

Reliability Framework

Objective

Transform sparsity information into an engineering metric.

Deliverables

Reliability Score

Reliability Categories

Reliability Visualizations

Outputs

reliability_scores.parquet

Required Validation

$0 \leq \text{Reliability} \leq 1$

Category Assignment Correct

No Missing Values

Completion Criteria

Reliability framework operational and validated.

Phase Gate

Reliability framework frozen before TKT engineering begins.

PHASE 8

Complete TKT Mutation Landscape

Objective

Generate and evaluate every possible single mutation in Transketolase.

Deliverables

Mutation Enumeration

Sparsity Calculations

Reliability Calculations

Stability Predictions

Outputs

tkt_single_mutation_landscape.parquet

tkt_mutation_analysis.parquet

Expected Scale

Length(TKT)

×

19 substitutions

for each residue.

Required Analysis

Reliability Landscape

Mutation Distribution

Candidate Distribution

Completion Criteria

Entire mutation space generated and analyzed.

Phase Gate

TKT landscape frozen before fitness scoring.

PHASE 9

Industrial Fitness Framework

Objective

Generate engineering candidate rankings.

Deliverables

Industrial Fitness Score

Candidate Ranking Engine

Top Candidate Tables

Outputs

industrial_fitness_scores.parquet

top_candidate_mutations.csv

Required Validation

Ranking Stable

Scores Normalized

No Missing Values

Output Sorted Correctly

Primary Output

Top Industrial Candidates

for future experimental evaluation.

Completion Criteria

Candidate ranking system fully operational.

Phase Gate

Engineering outputs frozen before publication generation.

PHASE 10

Figures, Publication Assets, And Science Fair Outputs

Objective

Generate all communication assets.

Deliverables

Publication Figures

Science Fair Figures

Presentation Figures

Summary Tables

Killer Figure

Required Figure Groups

Discovery Figures

Validation Figures

Integrated Analysis Figures

Reliability Figures

TKT Landscape Figures

Industrial Fitness Figures

Killer Figure

The final project figure must visually communicate:

Megascale-D



Combined Sparsity



Reliability Framework



Complete TKT Mutation Landscape



Industrial Fitness Ranking



Top Candidate Mutations

in a single visual.

Publication Package

Final outputs include:

Publication Tables

Publication Figures

Supplementary Material

Methods Documentation

Dataset Documentation

Science Fair Package

Final outputs include:

Poster Figures

Board Graphics

Judge Summary

Presentation Slides

Project Abstract

Final Project Acceptance Criteria

The project is complete when:

- ✓ Infrastructure Complete
 - ✓ Datasets Validated
 - ✓ Sparsity Framework Complete
 - ✓ Experiment 1 Complete
 - ✓ Experiment 2 Complete
 - ✓ Integrated Reliability Analysis Complete
 - ✓ Reliability Framework Complete
 - ✓ Full TKT Landscape Complete
 - ✓ Industrial Fitness Framework Complete
 - ✓ Candidate Rankings Generated
 - ✓ Publication Figures Generated
 - ✓ Killer Figure Generated
 - ✓ Science Fair Assets Complete
-

Project End State

The final system must answer three questions:

1.

Does mutation-space sparsity
predict prediction failure?

2.

Can prediction reliability
be estimated?

3.

Can reliability improve
protein-engineering decisions?

When those questions have been answered through a fully reproducible computational framework, the implementation is considered complete.

End Of Implementation Roadmap